

The Art of Solving Thermodynamics Problems

Why bother?

Engineering is puzzle-solving

- Innovation and design, at heart, are complex puzzles.
- A thermodynamics problem is a puzzle in miniature.
- Learn to solve the small one well, and the big ones get easier.

How you crack any puzzle

1. Learn the **rules of the game**.
2. Take stock of **starting information**.
3. Rules + information + logic
4. Deduce **new information**
5. Repeat until you solve

			9			1		2
7			3			6		
		2					3	
9					8	7		
3				1				9
		6	5					1
	1					4		
		4			9			6
8		3			6			

Thermodynamics is the dojo

- Its rules are genuinely intricate: they reward real understanding *and* disciplined, systematic use.
- Train in this dojo and you are ready for almost any engineering puzzle.

Planning is ~80% of the work

Plan first, compute second:

Rough worksheet → clean answer

- The thinking happens on the **rough worksheet**, before any numbers.
- Once the plan is set, the arithmetic is trivial bookkeeping.
- Dive in without a plan and you get lost — your clean answer sheet *becomes* your rough sheet.

The art of the doodle

- There is a craft to organising your thoughts as you scribble.
- That craft — **plan the strategy, then transcribe a clean response** — is what this series teaches.

What's ahead

- **Chunk 2:** the rules of the game (the five buckets) and a method to plan your attack.
- **Chunk 3:** the whole method, worked live on a real problem.
- Our playground: the **Perfect Ideal Gas**.